

1. (a) **Similarity:** The formula for the t -test (for testing $H_0 : \beta_1 = 0$) is:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}. \quad (1)$$

And the formula for the homoskedasticity-only F -test is:

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)}. \quad (2)$$

In both equations, the denominators play the role of standardization, and the numerator compares certain differences: in the t -test, the numerator **compares the difference between the estimator $\hat{\beta}_1$ and the hypothesized values 0**, and the null hypothesis is rejected when such difference is sufficiently large; in the F -test, the numerator **compares the difference between the fittings of the unrestricted and restricted models**, and the null hypothesis is rejected when such difference is sufficiently large (that is, the restricted model is rejected when the fitting of the unrestricted model is much better than that of the restricted model).

Disimilarity: there are many, for example, the t -test can only test for single restriction ($q = 1$), while the F -test can test multiple restrictions; the t -test can take negative values, while the F -test can only take non-negative values; in large samples the t -test follows $N(0, 1)$ under the null hypothesis, while in large samples the F -test follows $F_{q, \infty}$ (or χ_q^2/q) under the null hypothesis, etc.

- (b) For the linear regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + u_i, \quad i = 1, \dots, n, \quad (3)$$

often we want to test the joint hypothesis that all the slope coefficients of the regressors $X_{1i}, \dots, X_{ki}$ are zero. That is, the null and alternative hypotheses are

$$H_0 : \beta_1 = 0, \beta_2 = 0, \dots, \beta_k = 0 \text{ vs. } H_1 : \beta_j \neq 0, \text{ for at least one } j, j = 1, \dots, k.$$

In this case, the restricted regression model **only has an intercept** (since all the slope coefficients are zero), and **the corresponding R^2 will be zero**. Therefore, we have $R_r^2 = 0$. In addition, here we have $q = k$, the homoskedasticity-only F -statistic thus becomes

$$F = \frac{R_u^2/k}{(1 - R_u^2)/(n - k - 1)}, \quad (4)$$

where R_u^2 is calculated from the unrestricted regression model. (4) is called the “overall” regression F -statistic and it follows χ_k^2/k ($F_{k, \infty}$) when H_0 is true.

(c) Consider the following example:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{1i} \times D_{2i} + u_i, \quad i = 1, \dots, n, \quad (5)$$

where Y_i denotes individual i 's wage. D_{1i} and D_{2i} are dummy variables. $D_{1i} = 1$ if i has a bachelor's degree and $D_{1i} = 0$ otherwise. $D_{2i} = 1$ if i is female and $D_{2i} = 0$ otherwise. $D_{1i} \times D_{2i}$ is the interaction between the two dummy variable.

Then, for the population regression lines of the model, we can consider the following 4 scenarios with their corresponding parameters (coefficients):

- (1) $\{D_{1i} = 0, D_{2i} = 0\} = \{no\ bachelor, male\} : \beta_0$
- (2) $\{D_{1i} = 1, D_{2i} = 0\} = \{bachelor, male\} : \beta_0 + \beta_1$
- (3) $\{D_{1i} = 0, D_{2i} = 1\} = \{no\ bachelor, female\} : \beta_0 + \beta_2$
- (4) $\{D_{1i} = 1, D_{2i} = 1\} = \{bachelor, female\} : \beta_0 + \beta_1 + \beta_2 + \beta_3,$

Then, β_1 can be identified by (2) – (1), and β_2 can be identified by (3) – (1).

Finally, β_3 can be identified by using the difference-in-differences (DID) as follows:

$$\begin{aligned} & [(4) - (3)] - [(2) - (1)] \\ &= [(\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_2)] - [(\beta_0 + \beta_1) - \beta_0] \\ &= (\beta_1 + \beta_3) - \beta_1 = \beta_3. \end{aligned} \quad (6)$$

Notice that the interpretation of β_3 is

$$[\{bachelor, female\} - \{no\ bachelor, female\}] - [\{bachelor, male\} - \{no\ bachelor, male\}],$$

where $[\{bachelor, female\} - \{no\ bachelor, female\}]$ **fixes the gender as female and obtain the change in wage when one has a bachelor's degree (compared with those without a bachelor's degree).**

On the other hand, $[\{bachelor, male\} - \{no\ bachelor, male\}]$ **fixes the gender as male and obtain the change in wage when one has a bachelor's degree.**

Therefore, β_3 corresponds to the **extra effect (interaction effect)** of both having the bachelor's degree and being female. Intuitively, if β_3 is positive, then the employers may have a favouritism for female university graduates over male university graduates (thus an extra effect in wage, which is captured by β_3), and vice versa.

2. First, we collect the necessary information: $R_u^2 = 0.113$, $R_r^2 = 0.103$, $q = 2$, $k = 3$, and $n = 706$. In addition, when the null hypothesis is true, the homoskedasticity-only F-statistic follows the distribution $F_{2,\infty}$ (or $\chi_2^2/2$) in large samples because $q = 2$.

$$F = \frac{(R_u^2 - R_r^2)/q}{(1 - R_u^2)/(n - k - 1)} = \frac{(0.113 - 0.103)/2}{(1 - 0.113)/(706 - 4)} = 3.9572 > 3. \quad (7)$$

Here, 3 is the 5% critical value of $F_{2,\infty}$ (“ ∞ ” means that we are using large-sample approximation to the distribution of the F -test, just as we use large-sample approximation to the distribution of the t -test). Therefore, we can reject the joint null hypothesis at the 5% level because the value of the computed F -statistic is in the rejection region.

3. (a) i. For testing $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$,

$$|t| = \left| \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \right| = \left| \frac{0.352 - 0}{0.021} \right| = 16.7619 > 1.96. \quad (8)$$

Therefore, the high school earning difference is statistically significant at the 5% level. The 95% confidence interval for β_1 is

$$\begin{aligned} & \left[\hat{\beta}_1 - 1.96 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \cdot SE(\hat{\beta}_1) \right] \\ &= [0.352 - 1.96 \cdot 0.021, 0.352 + 1.96 \cdot 0.021] = [0.3108, 0.3932]. \end{aligned} \quad (9)$$

- ii. For testing $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 \neq 0$,

$$|t| = \left| \frac{\hat{\beta}_2 - 0}{SE(\hat{\beta}_2)} \right| = \left| \frac{0.458 - 0}{0.021} \right| = 21.8059 > 1.96. \quad (10)$$

Therefore, the male-female earning difference is statistically significant at the 5% level. The 95% confidence interval for β_2 is

$$\begin{aligned} & \left[\hat{\beta}_2 - 1.96 \cdot SE(\hat{\beta}_2), \hat{\beta}_2 + 1.96 \cdot SE(\hat{\beta}_2) \right] \\ &= [0.458 - 1.96 \cdot 0.021, 0.458 + 1.96 \cdot 0.021] = [0.4168, 0.4992]. \end{aligned} \quad (11)$$

- (b) i. For testing $H_0 : \beta_3 = 0$ vs. $H_1 : \beta_3 \neq 0$,

$$|t| = \left| \frac{\hat{\beta}_3 - 0}{SE(\hat{\beta}_3)} \right| = \left| \frac{0.011 - 0}{0.001} \right| = 11 > 1.96. \quad (12)$$

Therefore, the coefficient of age is statistically significant at the 5% level. The 95% confidence interval for β_3 is

$$\begin{aligned} & \left[\hat{\beta}_3 - 1.96 \cdot SE(\hat{\beta}_3), \hat{\beta}_3 + 1.96 \cdot SE(\hat{\beta}_3) \right] \\ &= [0.011 - 1.96 \cdot 0.001, 0.011 + 1.96 \cdot 0.001] = [0.009, 0.013], \end{aligned} \quad (13)$$

which does not contain zero. Yes, age is an important determinant of wage.

- ii. Note that in the question, $\Delta Age = 10$ (the difference in Age , where $\Delta Age = \Delta X_3$). Then, according to the given formula, the 95% confidence interval for $\beta_3 \Delta Age$ is

$$\begin{aligned} & \left[(\hat{\beta}_3 - 1.96 \cdot SE(\hat{\beta}_3)) \Delta Age, (\hat{\beta}_3 + 1.96 \cdot SE(\hat{\beta}_3)) \Delta Age \right] \\ &= [0.009 \cdot 10, 0.013 \cdot 10] = [0.09, 0.13]. \end{aligned} \quad (14)$$

(c) For column (3), the regression model can be written as:

$$AWE_i = \beta_0 + \beta_1 HighSchool_i + \beta_2 Male_i + \beta_3 Age_i + \beta_4 North_i + \beta_5 South_i + \beta_6 East_i + u_i, \quad (15)$$

where $i = 1, \dots, n$. To test if there is any important regional difference, we can form the following hypotheses:

$$H_0 : \beta_4 = \beta_5 = \beta_6 = 0, \text{ vs. } H_1 : \beta_j \neq 0 \text{ for at least one } j, j = 4, 5, 6. \quad (16)$$

The **unrestricted regression model** is (15), from which we can obtain R_u^2 . The **restricted regression model, which imposes** H_0 , can be written as:

$$AWE_i = \beta_0 + \beta_1 HighSchool_i + \beta_2 Male_i + \beta_3 Age_i + u_i, \quad (17)$$

where $i = 1, \dots, n$, and then we obtain R_r^2 from (17). Also, we note that $q = 3, k = 6$, and $n = 10973$. Therefore, we can define the homoskedasticity-only F -statistic as

$$F = \frac{(R_u^2 - R_r^2)/3}{(1 - R_u^2)/(10973 - 6 - 1)}. \quad (18)$$

According to the table of estimation results in the question, $R_u^2 = 0.0814$, $R_r^2 = 0.0761$, so that the homoskedasticity-only F -statistic is equal to 21.09; the heteroskedasticity-robust F -statistic (F -statistic for regional effects = 0 in the table) is equal to 21.87. Given that $q = 3$, the 5% critical value for $F_{3,\infty}$ ($\chi_3^2/3$) is 2.6. We have $F > 2.6$ so that H_0 is rejected at the 5% level (no matter using the homoskedasticity-only or the heteroskedasticity-robust F test). Therefore, we conclude that there exists significant regional difference in wages.

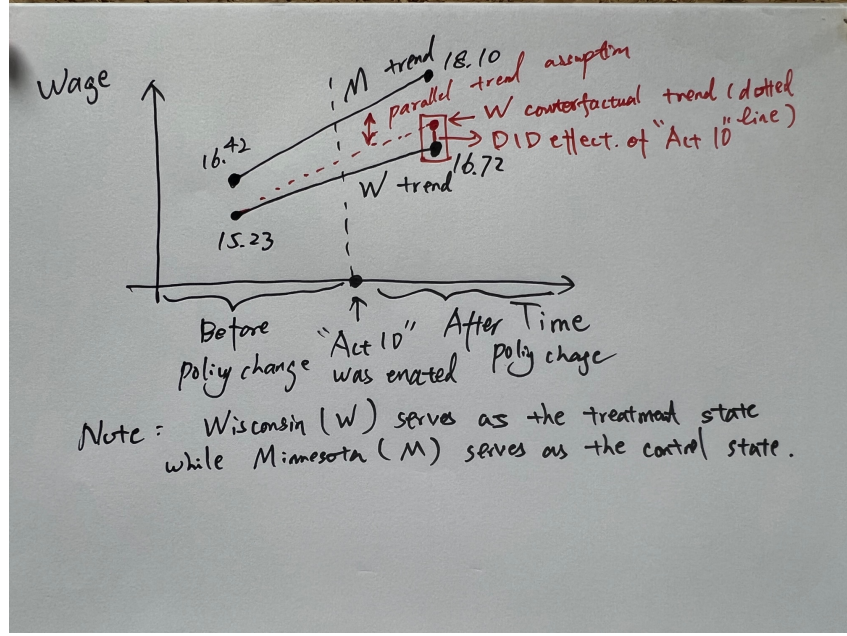
4. (a) According to the question, $(16.72 - 15.23) - (18.10 - 16.42)$ has the following meaning (let “avg.” denote “average”):

$$\begin{aligned} & (\text{avg.wage of Wisconsin after the legislation} - \text{avg.wage of Wisconsin before the legislation}) \\ & - (\text{avg.wage of Minnesota after the legislation} - \text{avg.wage of Minnesota before the legislation}), \end{aligned}$$

which can be seen as

$$\begin{aligned} & (\text{change in avg.wage due to US macroeconomy} + \text{the effect of the legislation}) \\ & - (\text{change in avg.wage due to US macroeconomy}) \\ & = \text{the effect of the legislation.} \end{aligned} \quad (19)$$

Therefore, the difference-in-differences (DID) estimate wipes out the effect due to the US macroeconomy so that we can find out what is the net effect of the Act 10 legislation. According to (19), **the key assumption is that during this time period, the US macroeconomy affected the average wage in Wisconsin and Minnesota in the same way (parallel trend assumption).**



- (b) Let us consider an example that studies the effect of free lunch program on academic performance (Gordanier et al., Economics of Education Review, 2020). The paper studies the effect of the Community Eligibility Provision (CEP), a free-lunch program, on elementary and middle school students' academic performance in South Carolina.

Treatment group: students from eligible schools that receive free lunches

Control group: students from schools that do not participate in the program

Before/After: before/after the introduction of the free lunch program

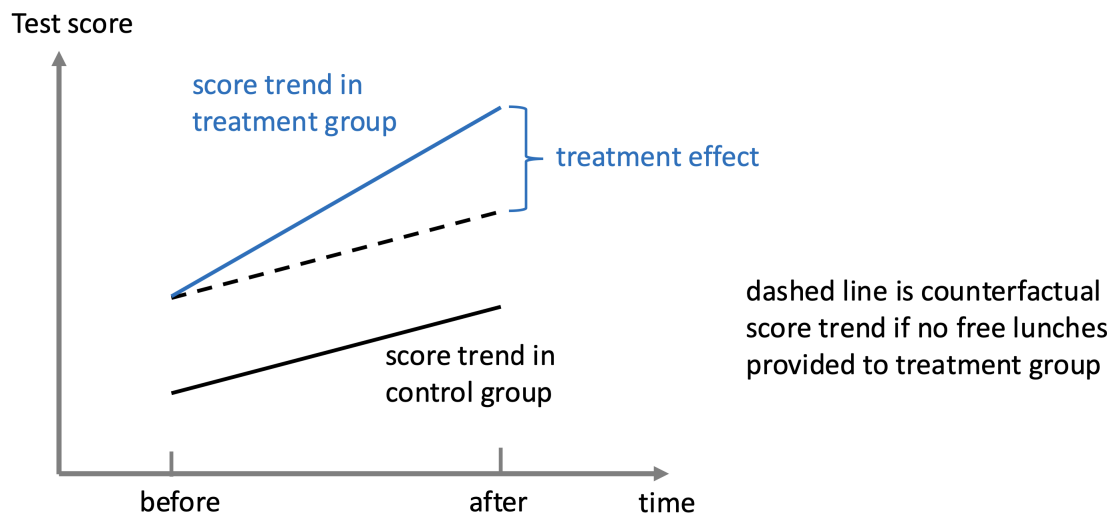
The outcome (dependent) variable is a measure of students' performance using math and reading test scores.

Denote $\bar{Y}$ as the average test score, we have the DID estimate of the treatment effect of free lunch program as follows:

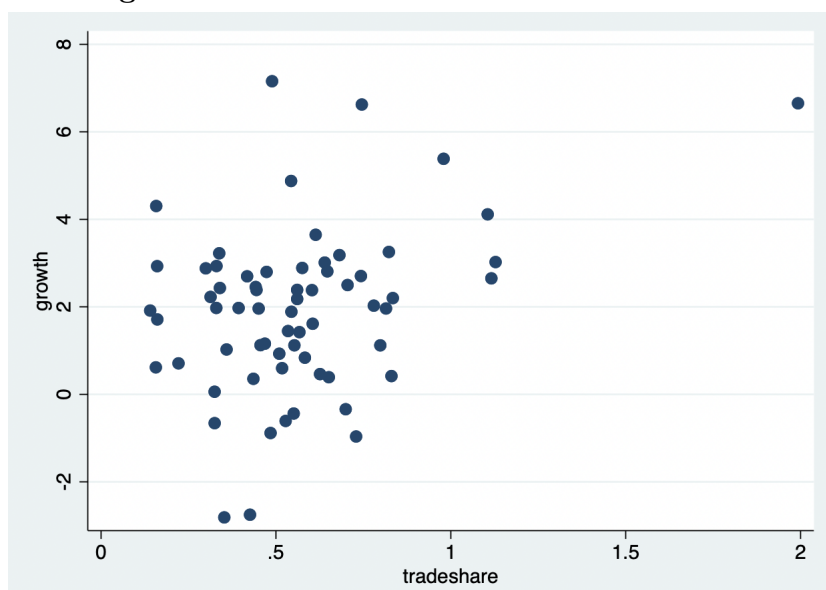
$$\begin{aligned} & \text{Treatment Effect} \\ &= (\bar{Y}_{treat,after} - \bar{Y}_{treat,before}) - (\bar{Y}_{control,after} - \bar{Y}_{control,before}), \end{aligned} \quad (20)$$

which can be seen as:

$$\begin{aligned} &= (\text{"change in students' performance due to learning process"} + \text{"the effect of free lunch"}) \\ &\quad - (\text{"change in students' performance due to learning process"}) \end{aligned} \quad (21)$$



5. (a) scatter growth tradeshare



As we can see in the graph there is one outlier *Malta*, which has much larger value of *tradeshare*. From OLS Assumption 3 we know that if there exists outliers, the OLS estimator will be biased. This can also be seen by comparing the OLS estimates with and without *Malta*:

reg growth tradeshare, robust gives the slope estimate 2.30, while

reg growth tradeshare if country name != "Malta", robust gives the slope estimate 1.68.

. reg growth tradeshare, robust

Linear regression	Number of obs	=	65
	F(1, 63)	=	12.09
	Prob > F	=	0.0009
	R-squared	=	0.1237
	Root MSE	=	1.79

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
tradeshare	2.306434	.6632868	3.48	0.001	.9809608	3.631907
_cons	.6402653	.4591457	1.39	0.168	-.2772641	1.557795

. reg growth tradeshare if country != "Malta", robust

Linear regression	Number of obs	=	64
	F(1, 62)	=	3.77
	Prob > F	=	0.0567
	R-squared	=	0.0447
	Root MSE	=	1.7894

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
tradeshare	1.680905	.8656171	1.94	0.057	-.0494392	3.411249
_cons	.9574107	.5360579	1.79	0.079	-.1141537	2.028975

(b) **reg growth tradeshare yearsschool rev_coups assassinations rgdp60, robust**

The coefficient for *yearsschool* has the value of t-test equal to 4.68 > 1.96 (and equivalently *p*-value smaller than 5%), thus significant at the 5% level. In addition, the fact that the 95% confidence interval does not include 0 also means that the value 0 is rejected by the t-test at the 5% level.

. reg growth tradeshare yearsschool rev_coups assassinations rgdp60, robust

Linear regression	Number of obs	=	65
	F(5, 59)	=	11.20
	Prob > F	=	0.0000
	R-squared	=	0.3589
	Root MSE	=	1.5821

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
tradeshare	1.561696	.584982	2.67	0.010	.3911494	2.732242
yearsschool	.5748461	.1228068	4.68	0.000	.3291101	.820582
rev_coups	-2.157503	.8782778	-2.46	0.017	-3.914933	-.400073
assassinations	.3540784	.3654816	0.97	0.337	-.3772485	1.085405
rgdp60	-.0004693	.0001192	-3.94	0.000	-.000708	-.0002307
_cons	.4897603	.7379584	0.66	0.509	-.9868911	1.966412

- (c) *Oil* is omitted by STATA because of **multicollinearity**. The OLS regression cannot be run if it were included. By checking the “Data Editor” in STATA, we can see that *Oil* is equal to 0 for all the observations.

	country_name	growth	oil	rgdp60	tradeshare	yearsschool	rev_coups	assasinations
1	India	1.915168	0	765.9998	.140502	1.45	.1333333	.8666667
2	Argentina	.6176451	0	4462.001	.156623	4.99	.9333333	1.933333
3	Japan	4.304759	0	2954	.1577032	6.71	0	.2
4	Brazil	2.930097	0	1784	.1604051	2.89	.1	.1
5	United States	1.712265	0	9895.004	.160815	8.66	0	.4333333
6	Bangladesh	.7082631	0	951.9998	.2214584	.79	.3064815	.175
7	Spain	2.880327	0	3123.001	.2994059	3.8	.0666667	1.433333
8	Colombia	2.227014	0	1684	.3130729	2.97	.1	.7666667
9	Peru	.060206	0	2019	.3246127	3.02	.2666667	.5666667
10	Haiti	-.6579341	0	923.9999	.3247456	.7	.3740741	.2

- (d) `reg growth tradeshare yearsschool rev_coups assassinations rgdp60, robust`
`test rev_coups assassinations`

As can be seen below, the p -value of the heteroskedasticity-robust F -test is equal to 0.0546. Therefore, the joint null hypothesis that none of the political/criminal factors has effect on economic growth rate can be rejected at the 10% level but not 5% level.

`. reg growth tradeshare yearsschool rev_coups assassinations rgdp60, robust`

Linear regression	Number of obs	=	65
	F(5, 59)	=	11.20
	Prob > F	=	0.0000
	R-squared	=	0.3589
	Root MSE	=	1.5821

growth	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
tradeshare	1.561696	.584982	2.67	0.010	.3911494	2.732242
yearsschool	.5748461	.1228068	4.68	0.000	.3291101	.820582
rev_coups	-2.157503	.8782778	-2.46	0.017	-3.914933	-.400073
assassinations	.3540784	.3654816	0.97	0.337	-.3772485	1.085405
rgdp60	-.0004693	.0001192	-3.94	0.000	-.000708	-.0002307
_cons	.4897603	.7379584	0.66	0.509	-.9868911	1.966412

`. test rev_coups assassinations`

(1) `rev_coups = 0`

(2) `assassinations = 0`

F(2, 59) =	3.06
Prob > F =	0.0546